

User Guide for Formula Subset Analysis

1. Download Data

a、 PubChem: For the search of compound mass formula candidates

(<https://ftp.ncbi.nlm.nih.gov/pubchem/Compound/CURRENT-Full/XML/>)

/pubchem/Compound/CURRENT-Full/XML

[父目録]

名称	大小	已修改日期
Compound_000000001_000025000.xml.gz	41.4 MB	2019/2/7 上午2:07:00
Compound_000025001_000050000.xml.gz	46.1 MB	2019/2/7 上午2:07:00
Compound_000050001_000075000.xml.gz	46.2 MB	2019/2/7 上午2:07:00
Compound_000075001_000100000.xml.gz	38.9 MB	2019/2/7 上午2:06:00
Compound_000100001_000125000.xml.gz	49.4 MB	2019/2/7 上午2:07:00
Compound_000125001_000150000.xml.gz	42.8 MB	2019/2/7 上午2:07:00
Compound_000150001_000175000.xml.gz	51.5 MB	2019/2/7 上午2:07:00
Compound_000175001_000200000.xml.gz	52.3 MB	2019/2/7 上午2:08:00
Compound_000200001_000225000.xml.gz	50.0 MB	2019/2/7 上午2:06:00
Compound_000225001_000250000.xml.gz	44.6 MB	2019/2/7 上午2:08:00
Compound_000250001_000275000.xml.gz	46.6 MB	2019/2/7 上午2:07:00
Compound_000275001_000300000.xml.gz	48.2 MB	2019/2/7 上午2:08:00
Compound_000300001_000325000.xml.gz	46.7 MB	2019/2/7 上午2:09:00
Compound_000325001_000350000.xml.gz	48.5 MB	2019/2/7 上午2:08:00
Compound_000350001_000375000.xml.gz	51.5 MB	2019/2/7 上午2:09:00

b、 HMDB Test dataset (<http://www.hmdb.ca/downloads>)

i. Compound information

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Search metabolites Search

Structures (in SDF Format)

Data Set	Released on	SDF File	File Size
Metabolite Structures	2018-07-09	Download	46.3 MB

Metabolite and Protein Data (in XML format)

Data Set	Released on	XML File	File Size
All Metabolites	2019-01-16	Download	614 MB
All Proteins	2019-01-14	Download	26.7 MB
Urine Metabolites	2019-01-17	Download	26.1 MB
Serum Metabolites	2019-01-17	Download	196 MB
CSF Metabolites	2019-01-17	Download	8.18 MB
Saliva Metabolites	2019-01-17	Download	16 MB
Feces Metabolites	2019-01-17	Download	61.6 MB
Sweat Metabolites	2019-01-17	Download	3.1 MB

ii. MS/MS spectra

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Saliva Metabolites	2019-01-17	Download	16 MB
Feces Metabolites	2019-01-17	Download	61.6 MB
Sweat Metabolites	2019-01-17	Download	3.1 MB

Spectra

Data Set	Released on	Download Link	File Size
Mass Spectra Image Files	2019-02-01	Download	166 MB
NMR Spectra FID Files	2019-02-01	Download	1.92 GB
Raw NMR Spectra Peaklist Files (TXT)	2019-02-01	Download	918 KB
Raw GC-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	23 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	158 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Experimental	2019-02-01	Download	2.28 MB
All Raw Spectra Peaklist Files (TXT)	2019-02-01	Download	1.68 GB
NMR Spectra Files (XML)	2019-02-01	Download	4.46 MB
GC-MS Spectra Files (XML) - Predicted	2019-02-01	Download	69 MB
GC-MS Spectra Files (XML) - Experimental	2019-02-01	Download	17.1 MB
MS-MS Spectra Files (XML) - Predicted	2019-02-01	Download	485 MB
MS-MS Spectra Files (XML) - Experimental	2019-02-01	Download	38.1 MB
All Spectra Files (XML)	2019-02-01	Download	4.1 GB

c 、 MassBank: MSFinder (Hydrogen rearrangement rules: computational MS/MS fragmentation and structure elucidation using MS-FINDER software. Analytical Chemistry 88, 7946-7958, 2016. <https://pubs.acs.org/doi/abs/10.1021/acs.analchem.6b00770> Table S5)

d 、 CASMI2016:

- i. Download Challenge Solutions Category 2+3 at <http://www.casmi-contest.org/2016/solutions-cat2+3.shtml>
- ii. Manually add the formula of the compounds to the last column.
- iii. Download the peak list file [Challenge_negative_peaklist.zip](#) for negative mode and [Challenge_positive_peaklist.zip](#) for positive mode at <http://www.casmi-contest.org/2016/challenges-cat2+3.shtml>

e 、 LipidBlast: <https://fiehnlab.ucdavis.edu/projects/lipidblast>, full release 3

2. Construct PubChem Dataset for Compound Formula Candidates Search

- a 、 Install MySQL Workbench and create a user "root" with password "1234" (MySQL Workbench Installation: A Step-by-Step Guide <https://www.simplilearn.com/tutorials/mysql-tutorial/mysql-workbench-installation>)
- b 、 Use the provided [readgzip_COMPOUNDtoMYSQL.py](#) to convert PubChem XML file into MySQL database table ('compounddata20190529')

3. Construct test datasets

a 、 HMDB

- i. Use the provided python file [HMDBtoMySQL4_0218.py](#) to convert HMDB Compound XML file into MySQL database table ('hmdbdata20190226')
- ii. Use the provided python files [MSMStoMySQL.py](#) and [MSMStoMySQL_otherproblem.py](#) to convert HMDB MS/MS XML file into MySQL database table ('msmsdata20190226')
- iii. Use the follow query to extract the qualified records

```
select t1.name, t1.accession, t1.chemical_formula, t1.monisotopic_molecular_weight,
      t2.ionization_mode, t2.instrument_type, t2.collision_energy_voltage, t2.peak_counter,
      t2.ms2_peak
from hmdbdata20190226 as t1
inner join msmsdata20190226 as t2
on t2.database_id=t1.accession
where (t2.instrument_type like '%LC-ESI-QTOF%' or t2.instrument_type like '%LC-ESI-ITFT%' or t2.instrument_type like '%LC-ESI-QFT%') and t2.peak_counter > 2 and
t1.chemical_formula like '%C%' and t1.chemical_formula like '%H%' and
t1.chemical_formula not like '%F%' and t1.chemical_formula not like '%Cl%' and
t1.chemical_formula not like '%Br%' and t1.chemical_formula not like '%I%' and
t1.chemical_formula not like '%B%';
```
- iv. Use the "Export" function in MySQL Workbench to save the resultant record to an Excel

Spreadsheet xml file "[FSA_HMDB_result_CHONSP.xml](#)"

- v. Open the file using MS Excel and save it as "[FSA_HMDB_result_CHONSP.xlsx](#)".

b 、 MassBank:

- i. Remove compounds with elements other than CHONSP.
- ii. Remove precursor types other than [M+H]⁺ and [M-H]⁻.
- iii. Save the resultant file as '[MSFINDER_MassBank_test_CHONSP.xlsx](#)'

c 、 CASMI2016:

- i. Remove compounds with elements other than CHONSP.
- ii. Remove precursor types other than [M+H]⁺ and [M-H]⁻.
- iii. Save the resultant file as '[solutions_casmi2016_cat2and3_CHONSP.csv](#)'

d 、 LipidBlast:

- i. Under the downloaded fold LipidBlast-Full-Release-3\LipidBlast-ASCII-spectra\, find [LipidBlast-neg.msp](#). Use the provided Matlab program "[LipidBlast_MS2_to_Table.m](#)" to convert the file into an MS Excel file "[LipidBlast_negative.xlsx](#)".
- ii. Under the downloaded fold LipidBlast-Full-Release-3\LipidBlast-ASCII-spectra\, find [LipidBlast-pos.msp](#). Use the provided Matlab program "[LipidBlast_MS2_to_Table.m](#)" to convert the file into an MS Excel file "[LipidBlast_positive.xlsx](#)".
- iii. Use MS Excel to merge the two files [LipidBlast_negative.xlsx](#) and [LipidBlast_positive.xlsx](#).
- iv. Remove compounds with elements other than CHONSP.
- v. Save the resultant file as '[LipidBlast_all_CHONSP.xlsx](#)'

4. Run FSA on the test dataset in Matlab (2021a or later version)

- a 、 FSA_HMDB.m (for the HMDB dataset)
- b 、 FSA_Massbank.m (for the MassBank dataset)
- c 、 FSM_CASMI2016.m (for the CSSMI2016 dataset)
- d 、 FSA_LipidBlast.m (for the LipidBlast dataset)